DGMS 2019 • SECTION H: ICHNOLOGY, REMOTE SENSING & GIS

Question 69: GPS Satellite Atomic Clocks: Rubidium & Cesium Standards

Satellite Trilateration Mechanics, Hyperfine Transitions, and Geodetic Timing Accuracies

OFFICIAL EXAMINATION QUESTION

GPS satellites carry rubidium and cesium atomic clocks that usually have time accuracy of 1 part in

(a) 10^2 and 10^3 respectively

(b) 10^2 and 10^8 respectively

(c) 10^6 and 10^{10} respectively

(d) **10^{12} and 10^{13} respectively [CORRECT]**

VERIFIED ANSWER KEY

Correct Option: (d) 10^{12} and 10^{13} respectively — In the Global Positioning System (GPS / NAVSTAR constellation), onboard space vehicles are equipped with ultra-stable atomic frequency standards: **Rubidium atomic clocks** ($^{87}\mathrm{Rb}$ gas cell) provide fractional stability of approximately **1 part in 10^{12}**, while **Cesium beam atomic clocks** ($^{133}\mathrm{Cs}$) achieve long-term accuracy of **1 part in 10^{13}** (and up to 10^{14}).

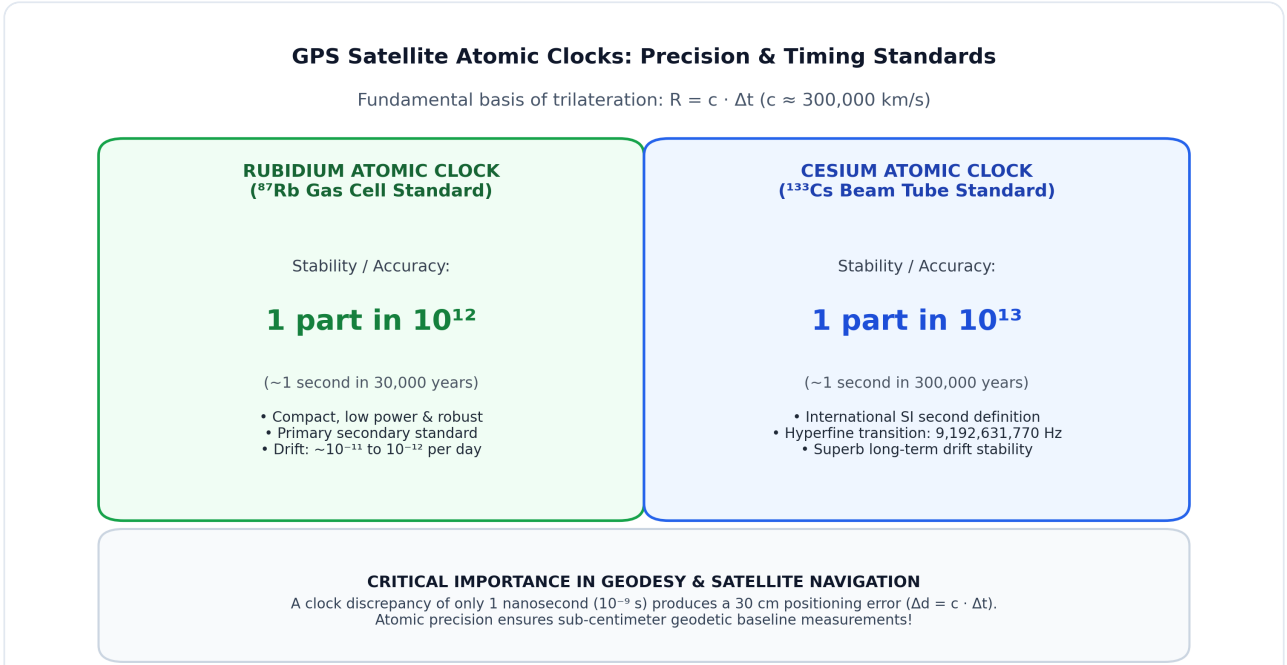


Figure 69.1 | Frequency Stability & Timing Standards of GPS Satellites: Comparison between onboard Rubidium standards (accuracy $\sim 1\text{ part in } 10^{12}$) and Cesium standards (accuracy $\sim 1\text{ part in } 10^{13}$). Because electromagnetic signals propagate at the speed of light ($c \approx 3 \times 10^8\text{ m/s}$), an infinitesimal clock error of just 1 nanosecond (10^{-9} s) produces a range error of ~30 cm; atomic stability is indispensable for sub-centimeter geodetic positioning.

1. Why Are Atomic Clocks Required in Satellite Geodesy?

The fundamental operating principle of the Global Positioning System (GPS) is **satellite trilateration**. A GPS receiver calculates its three-dimensional coordinates (X, Y, Z) on Earth by measuring the precise travel time (Δt) of radio signals transmitted from at least four satellites orbiting at an altitude of ~20,200 km:

$$\rho = c \cdot \Delta t = c \cdot (t_{\text{receive}} - t_{\text{transmit}})$$

where c is the speed of light ($299,792,458\text{ m/s}$).

- Because light travels ~30 cm in one nanosecond ($1\text{ ns} = 10^{-9}\text{ s}$), a timing error of only one microsecond (10^{-6} s) would cause a catastrophic positioning error of **300 meters**!
- To achieve meter-to-millimeter positioning accuracy needed for geological plate motion studies, surveying, and navigation, satellite clock drift must not exceed fractions of a nanosecond per day. This requires an oscillator stability of **1 part in 10^{12} to 10^{13}**.

2. Physics of Rubidium vs. Cesium Atomic Standards

Feature	**Rubidium Clock Standard ($^{87}\mathrm{Rb}$)**	**Cesium Beam Standard ($^{133}\mathrm{Cs}$)**
Frequency Standard	Ground-state hyperfine transition of Rubidium-87 vapor ($\approx 6.834682610\text{ GHz}$).	Hyperfine transition of Cesium-133 atom ($\equiv 9,192,631,770\text{ Hz}$, defining the SI second).
Time Accuracy / Stability	**1 part in 10^{12}** (loses ~1 second in 30,000 years)	**1 part in 10^{13}** (loses ~1 second in 300,000 to 3,000,000 years)
Short-Term Stability	Superior short-term Allan deviation ($1\text{ to }100\text{ seconds}$).	Moderate short-term stability; excellent medium-term.
Long-Term Drift	Exhibits slight optical pumping aging / drift over months.	Practically zero fundamental long-term drift (primary standard).
Hardware Attributes	Compact, lightweight, low power consumption.	Heavier, larger, higher power demand.
GPS Constellation Use	Block IIR, IIR-M, IIF, and GPS III satellites (primary workhorse).	Block II, IIA, and IIF satellites (historical reference).

3. Comprehensive Option Analysis

- **(a) 10^2 and 10^3 respectively:** *Incorrect.* 1 part in 10^2 (1%) or 10^3 (0.1%) corresponds to crude mechanical stopwatches, which lose minutes every day.
- **(b) 10^2 and 10^8 respectively:** *Incorrect.* 10^8 (1 part in 100 million) is typical of cheap quartz crystal oscillators (losing ~1 second every few months), totally unacceptable for satellite navigation.
- **(c) 10^6 and 10^{10} respectively:** *Incorrect.* 10^6 is standard wristwatch quartz; 10^{10} is temperature-compensated ovenized crystal (OCXO), still 100 to 1000 times too inaccurate for satellite clocks.

- **(d) 10^{12} and 10^{13} respectively: Correct.** Exact scientific values: Rubidium clocks achieve fractional stabilities of $\sim 1 \times 10^{-12}$ (1 part in 10^{12}), and Cesium clocks achieve $\sim 1 \times 10^{-13}$ (1 part in 10^{13}).

4. Authoritative Textbook References

- **Bernhard Hofmann-Wellenhof, H. Lichtenegger & E. Wasle (2008)** *GNSS – Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and more*, Springer, Chapter 5 ("Satellite Clocks and Orbits"), pp. 110–125.
- **G. Xu & Y. Xu (2016)** *GPS: Theory, Algorithms and Applications*, 3rd Ed., Springer, Chapter 2 ("GPS Signal and Clocks: Frequency Stability of Rubidium and Cesium Standards"), pp. 25–40.
- **Thomas M. Lillesand et al. (2015)** *Remote Sensing and Image Interpretation*, 7th Ed., Wiley, Chapter 1 ("Concepts and Foundations of Remote Sensing and GPS").
- **B. Bhatta (2021)** *Remote Sensing and GIS*, 3rd Ed., Oxford University Press, Chapter 9 ("Global Navigation Satellite System").